mass density of pure water at 4^0C. It is also defined as the ratio of specific weight of liquid to the specific weight of pure water at 4^0C. It is usually denoted by the symbol S. It has neither unit nor dimension. It is also called as relative density

Specific gravity/relative density

$$= \frac{\textit{mass density of liquid}}{\textit{mass density of pure water at } 4^\circ\text{C}}$$

$$S = \frac{\rho_l}{\rho_w}$$

Specific gravity/relative density

$$= \frac{\textit{specific weight of liquid}}{\textit{specific weight of pure water at } 4^\circ\text{C}}$$

$$S = \frac{\gamma_l}{\gamma_w}$$

4. Cohesion

Cohesion is the property of a liquid by which the molecules of the liquid are mutually attracted by each other. This property holds the molecules together.

Example: Mercury globules merge together and do not wet the surface of contact.

5. Adhesion

Adhesion is the property of a liquid by which the molecules of a liquid are attracted by the molecule of another kind of liquid or a solid. This property enables two different liquids adhere to each other or a liquid to adhere to solid body.

Example: Water drops adhere to the surface of contact rather than merging with each other.

6. Surface tension

Surface tension is a phenomenon in which the surface of a liquid, where the liquid is in contact with gas (such as air) acts like a thin elastic sheet. Surface tension is denoted by the Greek letter σ (sigma). It is expressed in Newton per metre i.e., N/m.

Examples:

i. **Floating a needle:** A carefully placed small needle can be made to float on the surface of water even though it is several times denser than water.

ii. **Walking on water:** Small insects such as the water strider can walk on water because their weight is not enough to penetrate the surface.

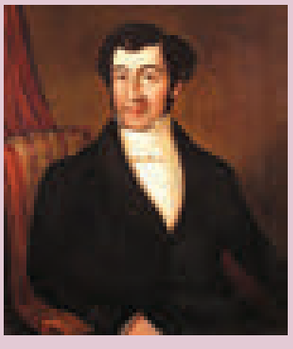
7. Capillarity

Capillarity is the rise or fall of a liquid in a capillary tube having small cross sectional

DO YOU KNOW?

Father of Hydraulics

Joseph Bramah



Born	13 April 1748 Stainborough, Barnsley, Yorkshire, England
Died	9 December 1814 (aged 66) Pimlico, London, England
Residence	London, England
Known for	Hydraulic press

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